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Validity of Four Consumer-Grade Optical Heart Rate Sensors for Assessing Volume and Intensity Distribution of Physical Activity

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ABSTRACT

Consumer-grade optical heart rate (HR) sensors emerged as promising tools to monitor volume and intensity of physical activity (PA). However, no validation study of optical HR sensors included recent comprehensive validation recommendations, required for facilitating usage for medical purposes. Validity of HR data measured with four consumer-grade optical sensors, the wrist-worn Garmin Venu 2S and Polar Vantage M2, and the upper arm-worn Polar Verity Sense and Scosche Rhythm24 were assessed in 32 participants over 24 h including various laboratory-based and free-living activities. Furthermore, validity of time at moderate or vigorous PA intensity zones was analyzed. A medical-grade ECG served as a reference. Reliability was assessed by analyzing data of laboratory-based activities during two visits. Across activities, the mean absolute percentage error ranged from 2.2% to 4.7% and intraclass correlation coefficients ranged from 0.91 to 0.98, indicating high validity for all optical sensors. All sensors validly detected time spent at moderate or vigorous intensities (mean error < 10%), except Polar M2 for moderate PA (mean error 12.8%) and Garmin V2 for vigorous PA (mean error −15.9%). Sensor day-to-day reliability was high, indicated by a mean absolute error of < 5 beats/min. Upper arm-worn sensors consistently outperformed wrist-worn sensors, particularly in activities involving increased arm movement and at higher intensities. Our findings identified an overall high validity of these four consumer-grade optical HR sensors, also for assessing time spent at higher PA intensities. However, differences were observed between sensors regarding activity subtypes and intensity levels. These data support informed decisions when selecting optical sensors for PA monitoring and intervention.

Trial Registration: The study was performed in compliance with the Declaration of Helsinki and its current amendments and was prospectively registered at ClinicalTrials.gov (ID NCT05525000).

1 | Introduction

Physical inactivity is a major modifiable risk factor for non-communicable diseases, which are the leading cause of death worldwide [1, 2]. Projections indicate that by 2030 about 499 million new cases of non-communicable diseases will occur if the amount of physical activity does not increase [3].

The World Health Organization currently recommends 150–300 min per week of moderate-intensity aerobic physical

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activity (MPA) or 75–150 min per week of vigorous-intensity physical activity (VPA), or an equivalent combination of moderate- and vigorous-intensity physical activity (MVPA) for substantial health benefits [4]. More recent studies advocate more personalized exercise intensity recommendations for additional health benefits, even for public health purposes [5]. This approach emphasizes the use of a polarized exercise intensity distribution and is favored by many endurance athletes. It is defined by a high volume spent at low-intensity exercise followed by a relatively large proportion spent at high intensity, with medium or moderate intensity clearly the smallest proportion [6].

In light of these advancements, it becomes crucial to precisely measure both the volume and intensity of PA throughout a 24 h period to even capture short bouts of moderate or vigorous PA [7, 8]. Wearable devices have emerged as powerful tools for continuously monitoring an individual's PA behavior and its popularity is increasing [9]. Among the sensors commonly integrated into these devices, accelerometers and optical HR sensors are the most common ones. Unlike accelerometers, optical HR sensors are particularly useful for prescribing and monitoring exercise based on physiological markers such as metabolic thresholds or maximum HR (HR_{max}) [10]. Unsurprisingly, optical HR sensors are increasingly used for applications in mobile health and telerehabilitation interventions [10].

However, it is important to note that most of these devices are marketed with general consumer electronics device safety marking (e.g., CE) intended for use as wellness/fitness devices only, since they lack the rigorous validation procedures when addressing a medical purpose, such as primary or secondary prevention of non-communicable diseases [11]. As a consequence, they are currently conservatively deployed by health experts in the medical domain [12]. Furthermore, recent scientific studies that validated optical HR sensors reveal a heterogeneous methodology with limited comparability and conflicting results, ranging from poor to good [13–15]. In addition, the product cycles from industry are much faster than the validation cycles. This highlights an acute need for comprehensive but also actionable validation standards [16]. Notably, comprehensive recommendations and standards for assessing the validity of consumer-grade optical HR sensors have recently been published [17, 18]. However, to date and to the best of our knowledge, there is no study that implemented these standards to evaluate the measurement quality of optical HR sensors. To this end, we aimed to fill this gap by investigating the validity and reliability of four widely used consumer-grade optical HR sensors using an electrocardiogram (ECG) as the reference criterion. The key focus of the investigation lay in evaluating the validity of cumulative amounts of PA, commonly referred to as activity minutes, and their distribution into respective exercise intensity zones. Further study aims were to investigate the validity of minimum HR (HR_{min})/HR_{max} measurement and user experience.

2 | Materials and Methods

2.1 | Study Design

In this prospective validation study, we equipped 32 participants with four consumer-grade photoplethysmography (PPG)-based optical HR sensors (serving as index measures) alongside a medical-grade ECG employed as the reference measure and monitored the participants for approximately 24 h. This period included two sessions of standardized laboratory activities, including cycling at different intensity levels, treadmill running, and simulated activities of daily living, as well as an ambulatory free-living period. The Ethics Committee of the State of Salzburg approved the study protocol and informed consent forms (No. 1017/2022). The study was performed in compliance with the Declaration of Helsinki and its current amendments [19].

2.2 | Participants

We invited healthy employees and patients with diverse underlying health conditions, all of whom were either currently attending or intending to attend a medically supervised exercise training program at the University Institute of Sports Medicine, Prevention and Rehabilitation of the University Hospital Salzburg to participate in the study. Inclusion criteria comprised an age of 35–80 years, while exclusion criteria encompassed signs of acute infections or diseases, atrial fibrillation, clinical signs of chronic venous insufficiency ≥ grade 1 at upper extremities, physical limitations for carrying out the planned examinations and tests, skin wounds or tattoos at sensors placement sites, or pregnancy.

Prior to data collection, informed consent was obtained from all participants. The sample size calculation was based on intraclass correlation coefficient (ICC), calculated according to the formula proposed by Bujang et al. (2017) [20]. With an acceptable agreement (R0) set at 0.90, an expected agreement (R1) at 0.95, and an alpha level at 0.05, the calculated sample size to achieve a power of 80% was 29 participants. To account for potential dropouts, we recruited 32 participants who completed the entire study protocol.

2.3 | Selection of Index and Reference Measures

The selection of consumer-grade optical HR sensors was guided by specific criteria, which were discussed within the study team and retrieved from manufacturers' specifications: CE marking, a battery life exceeding 24 h, a minimum HR sampling rate of 1 Hz, a data storage buffer of at least 24 h, and a cost below € 350. In addition, we focused on sensors that are widely used and available in the wearable market, furthermore limiting our selection to those that can be positioned on either the wrist or upper arm due to their ease of use and acceptance.

Following sensors were finally selected: Garmin Venu 2S (Garmin V2, Garmin Ltd., Olathe, Kansas, USA, firmware version 11.16, wrist-worn), Polar Verity Sense (Polar VS, Polar Electro Oy, Kempele, Finland, firmware version 1.1.5, wrist-worn), Polar Vantage M2 (Polar M2, Polar Electro Oy, Kempele, Finland, firmware version 2.0.12, upper arm-worn), and Scosche Rhythm24 (Scosche, Scosche Industries, Oxnard, California, USA, firmware version 6.4, upper arm-worn). Despite the Scosche having an internal memory duration of only 13 h, we chose to include it due to its widespread usage in

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upper arm-worn sensors. The reference criterion was a mobile 3-lead Holter ECG (ECG, Type: EP830, Amedtec ECGpro, Aue, Germany).

2.4 | Study Protocol

2.4.1 | Medical Examination

All study examinations were performed at the University Institute of Sports Medicine, Prevention and Rehabilitation, University Hospital Salzburg, Salzburg, Austria. A senior physician screened the participants for inclusion and exclusion criteria, including eligibility for maximal exercise testing. In the case of inclusion, skin type was assessed using the Fitzpatrick-Score [21].

2.4.2 | Sensor Placement

First, the 3-lead Holter ECG device was placed on the participant in accordance with recommendations from medical societies and the manufacturer's instructions [22]. Subsequently, the four optical HR sensors were placed according to the respective manufacturer's instructions. The placement of the Garmin V2 and the Polar M2 were randomized on either the dominant or the non-dominant wrist, while the Polar VS and Scosche were randomly positioned on the dominant and non-dominant upper arm, respectively. Notably, the Scosche sensor was applied during laboratory sessions only, due to its 13 h limited internal memory. Thus, free-living data do not include the Scosche device. In addition to the optical sensors and the Holter device, a 12-lead ECG (Amedtec ECGpro, Aue, Germany) was attached during all laboratory tests to evaluate electrocardiographic pathologies during exercise and to immediately identify HR_{max} after termination of the maximal treadmill running, which were subsequently used to derive personalized intensity zones during the following submaximal ergometer cycling. Figure 1 indicates sensor placement during a laboratory session.

2.4.3 | Laboratory Testing Day 1

2.4.3.1 | Activities of Daily Living. First, a laboratory activities of daily living (ADLs) protocol was performed in accordance with the expert statement of the INTERLIVE Network [17]. In addition, activities of daily living were selected based on recommendations from the Consumer Technology Association [18] as shown in Figure 2.

2.4.3.2 | Treadmill Running. Following the ADLs protocol, participants performed an incremental treadmill (Venus 200/75, h/p/cosmos sports & medical GmbH, Nussdorf-Traunstein, Germany) test using the modified Bruce protocol with stepwise increase of speed and slope [23]. Participants were free to choose whether to walk or run and were encouraged to exercise to voluntary exhaustion. A breath-by-breath metabolic analyzer (Metalyzer 3B, Cortex Biophysik GmbH, Leipzig, Germany) measured ventilation and gas exchange during the exercise test. Immediately after termination of the exercise test perceived exertion was assessed by the BORG scale [24]. Test termination was followed by a 5-min cool-down period at walking speed (3 km/h) on the treadmill.

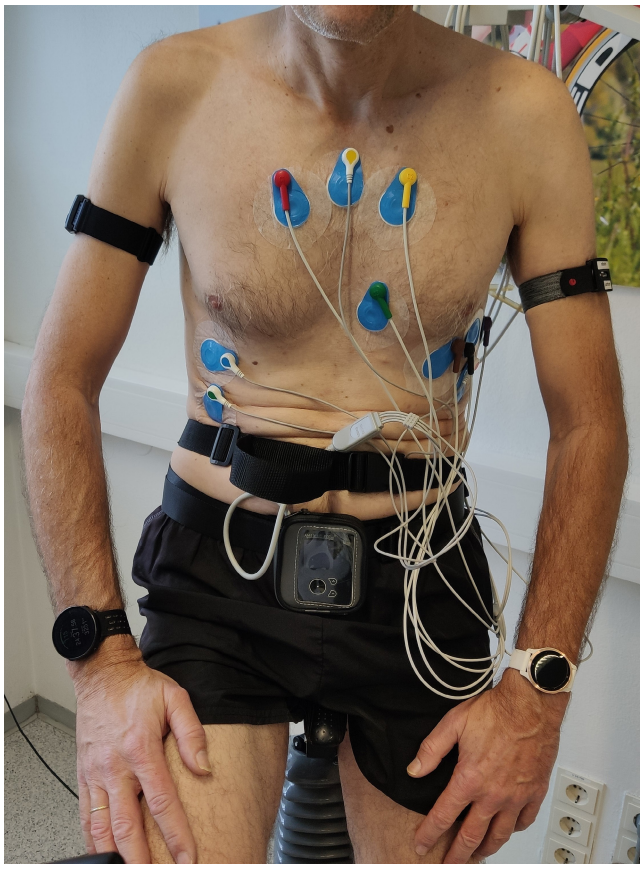

FIGURE 1 | Placement of optical HR sensors, 3-lead Holter ECG and 12-lead ECG during laboratory sessions.

2.4.3.3 | Determination of Intensity Zones. Based on the treadmill running test we determined HR_{max}. Therefrom we calculated the intensity zones according to current medical guidelines, as 57%–63%, 64%–76%, and 77%–95% of measured HR_{max} for light, moderate, and vigorous intensity, respectively [25].

2.4.3.4 | Ergometer Cycling. Participants remained seated quietly on the cycle ergometer until the HR dropped below the light-intensity zone. Then, exercise on the cycle ergometer (Ergoline, Ergoselect 200, Bitz, Germany) was performed as outlined in Figure 2.

2.4.4 | Free-Living and Laboratory Testing Day 2

After completion of the laboratory session on day 1, the 12-lead ECG was removed, and participants left the department with the 3-lead Holter ECG and the three optical HR sensors. They were instructed to resume their daily routines and return to the laboratory the next day at the same time as on day 1. During the laboratory session on day 2, the ADLs and ergometer cycling were repeated (Figure 2).

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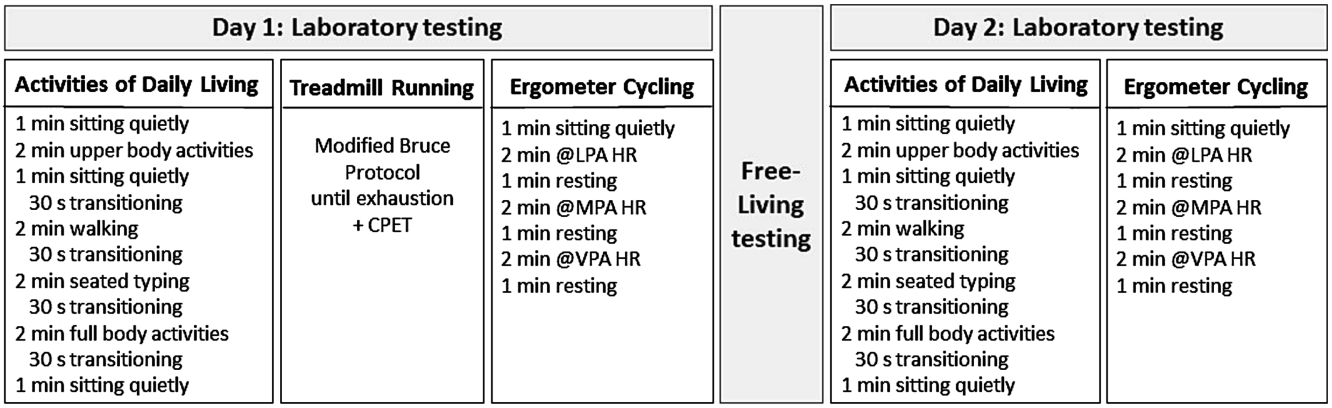

FIGURE 2 | Summary of all activities for validity and reliability assessment. Adapted from Mühlen JM et al., 2021 [17] and Consumer Technology Association, ANSI/CTA-2065, 2018 [18]. Duration of Laboratory testing day 1 approx. 2 h, duration of free-living testing approx. 22 h, and duration of laboratory testing day 2 approx. 1.5 h. Upper body activities consisted of washing the hands and taking shoes on and off; Full body activities consisted of stacking boxes, carrying a box, and sweeping the floor; CPET, Cardiopulmonary exercise testing; HR, Heart rate; LPA, Light physical activity; MPA, Moderate physical activity; VPA, Vigorous physical activity.

2.4.4.1 | Questionnaire. Finally, participants were asked to complete a questionnaire addressing their user experience with the sensors. The questionnaire was derived from the IDEA-Fast Technology Experience Questionnaire (ITEQ) [26].

2.5 | Data Collection and Processing

Before placement, each optical sensor was reset to factory settings. After completion of data collection, the Garmin V2 was connected to a desktop computer, and the .fit-file was downloaded. Similarly, both Polar sensors were connected to a desktop computer, all activity data were transferred to Polar Flow and the .csv-files were downloaded. As for the Scosche sensor, the companion app Rhythm SYNC (version 2.19) was used to download a .fit-file from the session. The .fit files from Garmin V2 and Scosche sensor were parsed using a Python package [27] to obtain the second-by-second timestamped HR data. The exported .csv-files for Polar sensors from Polar Flow were also timestamped based on the recording start time given in the file. No manufacturer disclosed the specific sampling frequency routine or the filters used despite further queries from the study team. HR data from the Holter-ECG device (sampling rate: 250 Hz; 12-bit resolution; transmission rate: 65 Hz after data compression; no filtering applied) were exported from ECGpro (Version 5.10.005, Amedtec ECGpro, Aue, Germany) as .qrs file and subsequently imported into GNU Octave (Copyright 1998–2021 John W. Eaton), where second-by-second timestamped ECG HR data were obtained. All the timestamps were converted from Coordinated Universal Time to local time. For further analysis of the time series dataset, a 10-s average of the second-by-second ECG HR data was employed.

2.6 | Data Cleaning

Three steps of data cleaning were implemented to obtain ECG HR data with the highest possible quality. First, we visually screened each ECG-recorded heartbeat for artifacts and undetected QRS complexes. If two or more heartbeats were falsely detected within a span of 10 heartbeats, we excluded it from further analyses due to its biasing effect on HR calculations. Second, we deleted all zero values after reporting their quantity. These occurrences of zero values were a result of the processing method applied to the ECG signal in GNU Octave. After step one and two, isolated artifact QRS-complexes remained. Therefore and third, we determined the HR_{min} and HR_{max} for each participant. HR_{min} was derived from the lowest HR recorded at nighttime, while HR_{max} was determined from the highest HR observed during maximal treadmill running. All ECG values outside of the interval (HR_{min}, HR_{max}) were rated implausible, and their quantity was reported and deleted. Implausibly high HR values in the ECG occurred primarily during periods of significant movement, particularly during the free-living phase and to some extent during treadmill running. In total, 1267 values exceeding HR_{max} were deleted, Whenever an ECG data point was deleted due to implausibility, the corresponding data from all four sensors were removed, too. Datapoints removed during the cleaning steps are summarized in Table S1.

2.7 | Synchronization

To assess validity and reliability of each optical sensor relative to the ECG serving as reference criterion, it is imperative to compare two continuous time series accurately aligned in time, as highlighted in previous studies [17, 28]. To address this issue, we cross-correlated the entire recording period on the per second timestamped cleaned dataset obtained from the previous steps [28]. In addition, a complementary method for time shift correction, namely RMSE (root-mean-square error), was applied for verification [28]. Considering that both methods yielded consistent results and based on recent recommendations [17], the cross-correlation method was selected for time-shift correction. For each participant the time shift with the highest cross-correlation value was applied (see Table S3). To ensure accuracy, the shifts were further validated visually, cross-referencing with HR landmarks such as HR_{max} during the maximal treadmill running.

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2.8 | Statistics

Descriptive statistics are given as arithmetic means ± standard deviation (SD) or numbers and percentages. Various statistical measures were computed to assess the validity and reliability of index optical HR sensors relative to the ECG. Data for validity testing was derived from laboratory testing day 1 and free-living period, whereas data for reliability testing was derived from laboratory testing day 1 and day 2.

Statistical measures for validity of HR included mean absolute error (MAE, expressed as mean absolute difference of all data points between reference and index measure) [29], mean absolute percent error (MAPE, expressed as the mean absolute percentage error between the reference and index sensor across all recorded seconds) [17, 18], Pearson correlation coefficient (r), coefficient of variation (CV, defined as the ratio of standard deviation divided by the mean) [30], and the two-way mixed model, absolute agreement intraclass correlation coefficient (ICC) [31]. Bland–Altman plots were employed to provide a graphical representation of the agreement and potential systematic biases between the reference and index measures [17]. Validity for HR measurements was considered high if the MAPE was ≤ 5% [13]. The classification of validity based on ICC involved four thresholds: poor < 0.50, moderate 0.50–0.75, good 0.75–0.90, or excellent > 0.90 [31].

Statistical measures for validity of PA volume spent at different intensities included mean error (expressed as mean difference between activity minutes from reference and index measure) and presented as absolute values and percentages. Validity for classifying activity minutes to intensity zones light physical activity (LPA), MPA, VPA, and MVPA was considered high if the mean percentage error was less than 10% (for justification see discussion).

Validity for detecting HR_{min} and HR_{max} by the optical sensors was considered high if the mean error between reference and index measure was ≤ 3 bpm [32].

Reliability or within-sensor variability was evaluated using HR data from laboratory day 1 and day 2 of ADLs and cycle ergometer sessions. This evaluation included calculating the mean error between the index sensor and the reference sensor on both days separately and subsequently calculating the MAE between day 1 and day 2 [33].

Statistical significance was set at $p < 0.05$. All analyses were conducted using IBM SPSS (Version 29.0.1.0, Armonk, NY, IBM Corp).

3 | Results

3.1 | Participants

A total of 32 subjects participated in the study and their baseline characteristics are outlined in Table 1. Participants' age ranged from 38 to 74 years. Among participants enrolled or scheduled for facility-based medically supervised exercise training, there were 12 healthy subjects and 14 patients with various cardiovascular conditions, including coronary artery disease ($n = 11$), chronic heart failure ($n = 1$), valvular heart disease ($n = 1$), and congenital heart disease ($n = 1$). Additionally, one participant had a metabolic disease, one had an oncologic disease, and four suffered from a post-COVID-19 condition. Eleven participants were taking HR limiting medication (10 taking Beta Blockers, 1 taking Ivabradine). The median score of the skin color tone on the Fitzpatrick Scale was 3 (20 participants), with 81% (26 participants) with lighter skin (Fitzpatrick 1–3) and 19% (6 participants) with darker skin (Fitzpatrick 4–6). All participants successfully completed the laboratory sessions on day 1, the free-living session, and the laboratory sessions on day 2.

TABLE 1 | Baseline characteristics of ($N = 32$ participants, 11 female).

	Mean ± SD	Min/Max
Age (years)	59.1 ± 10.1	38/74
Body height (cm)	173.3 ± 7.8	159.5/191.0
Body mass (kg)	79.0 ± 16.5	50.0/118.0
BMI (kg/m^2)	26.1 ± 4.4	19.5/36.4
HR_{min} (beats/min)	50.7 ± 7.5	37/69
HR_{max} (beats/min)	156.7 ± 20.3	122/190
$\dot{V}O_{2max}$ (mL/kg/min)	32.8 ± 7.8	17.5/52.2

Note: Data as mean ± SD, n, Min, and Max.
Abbreviations: $\dot{V}O_{2max}$, maximum oxygen uptake; BMI, Body Mass Index; HR_{max}, maximum heart rate; HR_{min}, minimum heart rate; Max, maximum value; Min, minimum value.

3.2 | Description of Data Set

After removing artifacts and implausible values from the ECG on average 94.1% of the initially recorded ECG data set remained. After removing missing data from the optical sensors, data availability was as follows: 99.1% for Garmin V2, 99.5% for Polar VS, 98.5% for Polar M2, and 100% for Scosche. With Scosche, one session of one participant was not recorded for unknown reasons, and consequently, all statistics for this sensor are based on 31 participants. Descriptive statistics of missing values of each sensor are summarized in Table S2.

3.3 | Validity of Optical-Based Heart Rate

The validity of HR between the optical sensors compared to the ECG based on MAPEs across all activities is illustrated in Figure 3. More comprehensive statistical measures can be found in the tables S4–S6. Overall, each optical sensor remained below the predefined validity cut-off of ≤ 5% MAPE, with values ranging from 2.2% (Polar VS) to 4.7% (Polar M2). Additionally, they demonstrated high ICCs, ranging from 0.91 to 0.98. Upper arm-worn sensors outperformed wrist-worn sensors consistently during all five activities (Figure 3). However, when evaluating each activity individually, notable differences in validity became apparent among the sensors. During ADLs MAPE for both wrist-sensors, Garmin V2 (5.6%; ICC: 0.88) and Polar M2 (7.2%; ICC: 0.80) were higher

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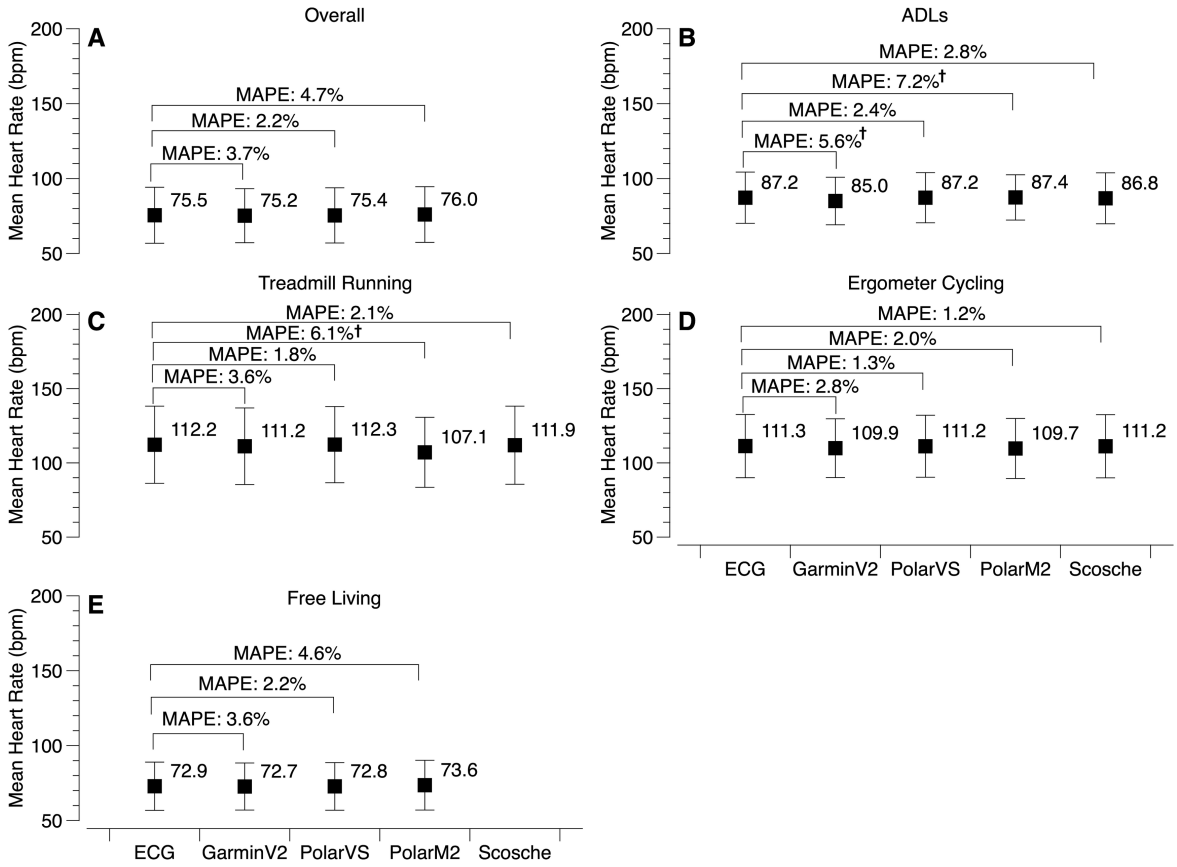

FIGURE 3 | Validity of optical HR sensors. HR (arithmetic mean ± SD) of electrocardiogram and optical sensors for Overall (Panel A: Encompassing laboratory testing day 1 and free-living period), ADLs (Panel B), Treadmill Running (Panel C), Ergometer Cycling (Panel D), and Free Living (Panel E). Validity is considered high, if MAPE is below 5%, with † indicating a MAPE ≥ 5%. MAPE, mean absolute percentage error; bpm, beats per min.

than 5%. During treadmill running, only Polar M2 exceeded the MAPE cut-off value of 5% (6.1%; ICC: 0.86). For the free-living phase, for the entire ergometer cycling as well as for each intensity on the cycle ergometer, all optical sensors remained below 5% MAPE (see Figure 3 and Table S5). Further, the MAPE of the upper arm-worn optical sensors decreased with increasing work intensity on the cycle ergometer, indicating higher validity at high-intensity zones compared to wrist-worn optical HR sensors (see Table S5). The MAE ranged from 1.0 for Scosche at ergometer cycling (moderate and vigorous intensity) to 7.0 for Polar M2 at treadmill running (see Tables S5 and S6).

The Bland–Altman plots for all laboratory activities on day 1, encompassing ADLs, treadmill running, and ergometer cycling are shown in Figure 4. While the systematic error remained below 2 bpm for all optical sensors, it was more than four-fold higher for wrist-worn sensors compared to upper arm-worn sensors. Additionally, the limits of agreement were twice as wide for wrist-worn sensors compared to their upper arm-worn. Bland–Altman plots for activity minutes (min) at different PA intensities across all activities are shown in Figures S1–S4.

3.4 | Validity of Detecting Physical Activity Volume at Different Intensities

Figure 5 illustrates the total activity minutes of each optical sensor and the ECG separated for the following intensity zones: LPA, MPA, VPA, and MVPA on laboratory activities on day 1 and the free-living period. All four optical sensors remained below the predefined cut-off of 10% mean percentage error when compared to the ECG, except Polar M2 (12.8%, +15.3 min) for MPA and Garmin V2 (−15.9%, −5.5 min) for VPA. The Polar VS showed less than half the error as the wrist-worn sensors with regard to the validity of activity minutes within MPA and MVPA intensity zones. The Polar M2, however, was more accurate than the Polar VS sensor (see Figure 5) in the VPA zone. The validity of the Scosche was high in each intensity zone, with a maximum error of 2.6% for VPA (see Figure 5), notably being based on laboratory recordings only.

3.5 | Validity of HR_{min} and HR_{max} Detection

Table 2 presents the validity of the optical sensors in detecting HR_{min} and HR_{max}. With regard to HR_{min}, Garmin V2 and Polar VS exhibited a mean error to the ECG of less than 3 bpm, while Polar M2 showed a mean error more than twice as high as −7.2 bpm. The validity in detecting HR_{max} generally exceeded that of HR_{min}, showing a mean error of no more than 2.8 bpm. The Polar M2 exhibited a considerably large error for HR_{max} (−10.9 bpm). Polar VS stood out by detecting HR_{min} and HR_{max} with an error ≤ 3 bpm for more than 80% of all participants. In contrast, Garmin V2, Polar M2, and Scosche achieved an error of less than or equal to 3 bpm in 45%–59% for HR_{min} and 53%–68% for HR_{max}.

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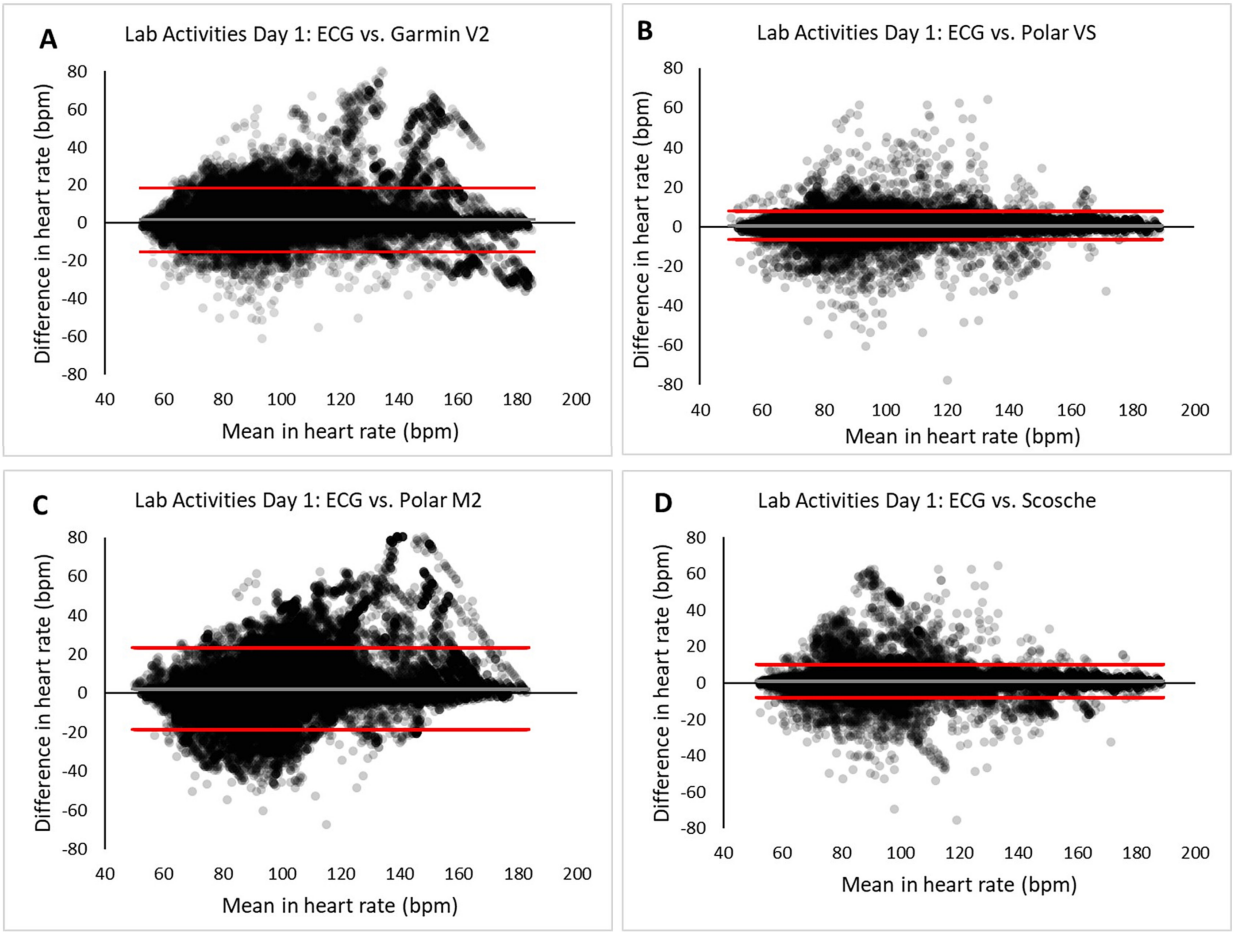

FIGURE 4 | Bland–Altman plots of HR for all lab activities on day 1. Each sensor (Panel A, Garmin V2; Panel B, Polar VS; Panel C, Polar M2; Panel D, Scosche) is compared to the electrocardiogram. The gray line indicates the systematic error (bias), and the red lines indicate 95% limits of agreement.

3.6 | Reliability

Table 3 shows mean error between sensors and ECG and MAE between mean error on day 1 and day 2 indicating day-to-day reliability of ADLs and ergometer cycling. Both upper arm-worn sensors, Polar VS and Scosche, demonstrated higher reliability than the wrist-worn sensors, exhibiting a MAE of ≤1 bpm. In contrast, wrist-worn optical sensors Polar M2 and Garmin V2 had a wider range of MAE from 1.5 bpm to 4.6 bpm, indicating a lower reliability.

3.7 | User Experience

Participants rated all optical sensors similarly in terms of comfortability. With regard to suitability for everyday use and visual appeal, both wrist-worn sensors achieved higher scores compared to their upper arm-worn sensors. This is also reflected in the overall evaluation, where wrist-worn sensors scored better than upper arm-worn sensors (see Table 4).

4 | Discussion

The main purpose of this study was to evaluate the validity and reliability of four consumer-grade optical HR sensors, acknowledging and adhering to contemporary and comprehensive validation standards that include the measurement of various daily activities and exercise routines over a 24 h period [17, 18]. Additional study aims were assessing the validity of activity minutes and their distribution across PA intensity zones, investigating the validity of HR_{min}/HR_{max} measurement, and obtaining data on user experience. Our findings suggest that the measurement quality of optical HR sensors and the detection of moderate- and vigorous-intensity PA volume were generally valid and reliable. Nevertheless, relevant differences were observed between the optical sensors investigated and across activity subtypes, with upper arm-worn sensors consistently outperforming wrist-worn sensors, particularly in activities involving increased arm movement.

To the best of our knowledge, our validation study on optical HR sensors is the first taking into account the latest validation recommendations [17]. Notably, these standards recommend utilizing MAPE as the primary statistical measure for assessing the validity of optical HR sensors [17]. While no specific cut-off is explicitly provided by the INTERLIVE Network, it indicates that acceptable validity expressed by MAPE may vary depending on the type of activity and intensity. In previous studies, MAPE cut-offs for defining validity range from 3%, particularly when validity is crucial, as in research scenarios, to 20%, often without

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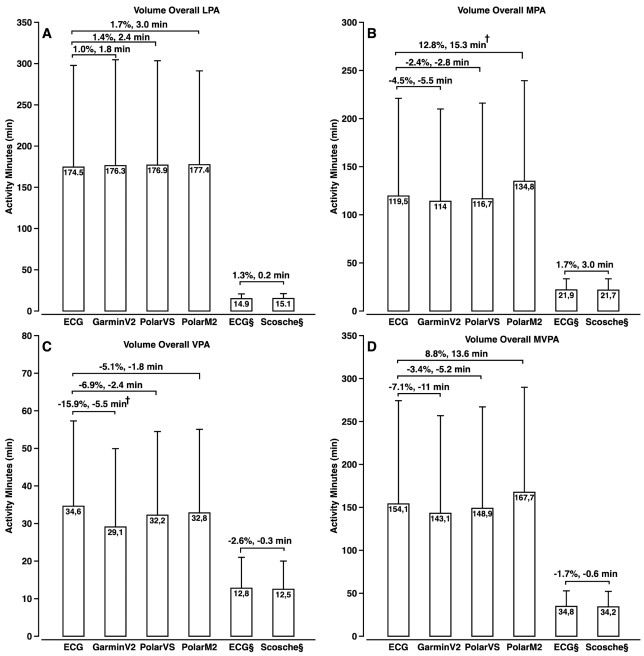

FIGURE 5 | Volume of activity minutes each sensor recorded for the intensity zones light (Panel A, LPA), moderate (Panel B, MPA), vigorous (Panel C, VPA), and moderate to vigorous physical activity (Panel D, MVPA) compared to the ECG for Overall, encompassing laboratory testing day 1 and free-living period. Validity is considered high, if the mean percentage error is below 10%, with † indicating a mean percentage error of more than 10%. §, Scosche has less data points since it was not used during the free-living phase due to its limited internal memory.

clear justification for chosen values [13, 15, 34]. In our study, we opted for a conservative MAPE cut-off ≤5%. This choice is not only conservative compared to other research groups but is also grounded in physiological and practical reasoning specific to utilizing optical sensors for PA monitoring and exercise prescription: Our diverse study population, which includes participants with underlying cardiovascular diseases taking HR limiting medication, had an average HR_{max} of 157 bpm. Applying current recommendations based on percentages of HR_{max}, the following intensity zones were calculated [25]: light intensity from 89 to 99 bpm, moderate intensity from 100 to 119 bpm, and vigorous intensity from 120 to 149 bpm. For exercise prescription, HR zones are typically recommended as a range of 10 heartbeats around the midpoint of the desired intensity zone. Accepting a MAPE of 10% for optical HR sensors, the measurement error would exceed the limits of the HR zones indicated above. In contrast, a MAPE of 5% would only slightly and negligibly exceed the HR limits at light intensity (84 bpm and 104 bpm) and the upper limit of moderate intensity (121 bpm). Based on this calculation, we conclude that, irrespective of the setting, whether in a laboratory or free-living environment, a MAPE of ≤5% can be considered acceptable to classify a sensor as valid, particularly when utilized for PA monitoring and exercise prescription.

Based on the cut-off value of 5% MAPE, all optical sensors of this study can be classified as valid when taking all activities into account. Nonetheless, noteworthy differences were observed between sensors and activities. Both wrist-worn sensors, Garmin

TABLE 2 | Validity for detecting HR_{min} and HR_{max} between ECG and each sensor.

	HR_{min}					HR_{max}				
	ECG [bpm]	Sensor [bpm]	*p*	Error [bpm]	≤3 beats difference	ECG [bpm]	Sensor [bpm]	*p*	Error [bpm]	≤3 beats difference
Garmin V2	51.7 ± 7.5	48.8 ± 7.0	0.114	−2.9 ± 2.4	19 (59%)	155.3 ± 19.9	157.1 ± 21.9	0.734	1.8 ± 11.3	19 (59%)
Polar VS	51.7 ± 7.5	49.0 ± 8.5	0.195	−2.6 ± 5.8	28 (88%)	155.3 ± 19.9	155.4 ± 21.3	0.98	0.1 ± 2.6	27 (84%)
Polar M2	51.7 ± 7.5	44.5 ± 6.6	< 0.001	−7.2 ± 7.9	17 (53%)	155.3 ± 19.9	149.0 ± 19.5	0.211	−10.9 ± 32.5	17 (53%)
Scosche[a]	57.6 ± 8.6	61.1 ± 8.7	0.115	3.5 ± 7.7	14 (45%)	155.8 ± 20.1	158.6 ± 20.6	0.584	2.8 ± 4.8	21 (68%)

Note: Mean HR ± SD and mean error ± SD for all sensors. Number and percent of measurements detected HR_{min} and HR_{max} with ≤3 beats difference.
[a]Data is based on 31 participants. Scosche has different HR_{min} and HR_{max} values since it was not used during the free-living phase due to its limited internal memory.

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TABLE 3 | Day-to-day reliability of each sensor. Agreement of optical sensors with ECG between day 1 and day 2.

	ADLs				Ergometer cycling			
	Garmin V2	Polar VS	Polar M2	Scosche	Garmin V2	Polar VS	Polar M2	Scosche
Mean error day 1, bpm	−2.2 (2.8)	0.0 (0.3)	1.2 (5.0)	−0.3 (1.8)	−1.6 (10.6)	−0.1 (0.4)	−0.7 (2.8)	−0.2 (0.3)
Mean error day 2, bpm	−2.3 (3.1)	0.1 (0.4)	2.1 (6.8)	−0.1 (0.9)	−2.5 (10.6)	−0.1 (0.5)	−0.3 (3.2)	−0.9 (3.7)
MAE, bpm (95% CI)	2.6 [1.8, 3.4]	0.3 [0.2, 0.4]	3.8 [1.9, 5.6]	1.0 [0.4, 1.6]	4.6 [2.3, 6.9]	0.4 [0.3, 0.5]	1.5 [0.7, 2.2]	0.3 [0.2, 0.4]

Note: Mean errors as the differences between ECG and sensor are shown as mean and SD.
Abbreviations: ADLs, Activities of daily living; Bpm, beats per min; MAE, Mean absolute error between Mean error on day 1 and day 2.

TABLE 4 | Questionnaire results for user experience.

	Q1: ...uncomfortable (1–10)	Q2: ...comfortable (1–5)	Q3: ...suitable for everyday use (1–5)	Q4: ...visually appealing (1–5)	Q5: Overall evaluation of the sensor (1–7)
Garmin V2	1.6 (1.7)	4.3 (1.1)	3.9 (1.3)	3.8 (1.5)	5.9 (1.4)
Polar VS	1.8 (2.0)	4.1 (1.3)	3.2 (1.6)	2.9 (1.3)	5.4 (1.3)
Polar M2	1.6 (1.1)	4.0 (1.2)	3.8 (1.1)	3.9 (1.3)	5.6 (1.4)
Scosche	1.1 (0.4)	3.8 (1.2)	3.0 (1.5)	3.0 (1.4)	5.0 (1.3)

Note: All data are arithmetic mean ± SD. Q1–4: The sensor was..., Scale Q1: 1 = low, 10 = high, Scale Q2–Q4: 1 = do not agree, 5 = agree; Scale Q5: 1 = negative, 7 = positive; Q1–5, Question 1–5.

V2 and Polar M2, showed a MAPE greater than 5% for ADLs. Similar results were observed for treadmill running, where wrist-worn sensors showed higher MAPE values (Garmin V2: MAPE 3.6% and Polar M2: MAPE 6.1%) than upper arm-worn sensors (Polar VS: MAPE: 1.8% and Scosche: MAPE 2.1%), with Polar M2 exceeding the 5% cut-off value. Upper arm-worn sensors showed higher validity compared to wrist-worn sensors for all five activities depicted in Figure 3. Noteworthy, the ADLs and the treadmill running involved tasks that required hand movements, and thus, the validity of the sensors may have been influenced by motion artifacts [17, 35]. Previous findings indicate that wrist-worn sensors exhibit higher validity during steady-state intensities such as ergometer cycling compared to activities with changing intensities like graded incremental tests or alternating intensities during ADLs [35, 36].

Another focus of this study was to assess the validity of optical HR sensors for evaluation of activity minutes and their distribution across PA intensity zones encompassing laboratory testing day 1 and free-living period. Therefore, it is important that the sensors allow for correctly categorizing the activity minutes into the appropriate intensity zone. Of note, we used the raw HR data from the sensors for PA intensity categorization instead of those provided from the manufacturers calculated by proprietary algorithms, considering the study aim was to validate HR data and since the algorithms are updated frequently. For this purpose, we used a maximum mean percentage error of 10% between any optical sensor and the ECG as a criterion for high validity. The rationale behind the threshold of 10% is based on the WHO recommendations of 150–300 min of MVPA per week [4]. Taking into account that in a rather physically inactive population performing the lower level of recommended PA is associated with the greatest health benefits, particularly for women [37], and additional health benefits decrease with increasing activity minutes [38], high validity at lower levels of PA is crucial. When performing 150 min of PA during a week, an error of 10% would translate to 15 min, which has been demonstrated to be acceptable, since the error would not impact endpoints such as all-cause mortality [37]. Thus, we deem 10% as an acceptable error across different PA volumes. Our study demonstrates that all four sensors are sufficiently valid for detecting activity minutes spent at MVPA over 24 h, with a mean error ranging from −1.7% to 8.8% compared to the ECG.

Notably, for patients with cardiovascular diseases as well as in the general population, it is also important to validly distinguish between activity minutes spent in MPA and VPA [5, 7] since the activity minutes spent in the VPA intensity zone are associated with additional health benefits [39]. Furthermore, recent studies indicate that even short bouts of VPA lasting less than 1 min may provide additional health benefits for previously inactive adults [8]. All optical sensors evaluated in our study detected MPA minutes with acceptable validity, except for the Polar M2, which exhibited a mean error of 12.8%, surpassing the 10% threshold compared to the ECG, indicating that it cannot be considered valid enough for accurate MPA time detection. For monitoring volume of VPA all sensors were suitable, except for Garmin V2, which exhibited a mean percentage error of −15.9%, also surpassing the 10% threshold compared to the ECG. Thus, if aiming for monitoring or prescription of vigorous exercise intensity, Garmin V2 cannot be considered valid for VPA time detection. Despite the Scosche sensor being valid in detecting intensity minutes, the limited

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memory of 13 h is not suitable for continuous monitoring or generating a weekly profile of PA volume, which may represent a clear limitation in certain settings. Overall, Polar VS stood out as the most valid sensor for 24 h PA monitoring and PA intensity allocation.

Many optical HR sensors and smart wearables use HR_{min} and average resting HR as another key metric for health [12]. Additionally, valid HR_{max} detection enables personalized PA intensity recommendations [25]. Thus, as add-on for healthy athletes, it could be a time and money saving method to validly measure HR_{max} with optical HR sensors within a strenuous, all-out high-intensity session. In addition, a falsely high HR_{max} is often a cause for concern in patients with cardiovascular diseases and athletes [40]. Therefore, we chose a conservative absolute error threshold of ±3 bpm compared to the ECG as cut-off value, as proposed by literature [29] and in addition being below ±3 bpm in 80% of participants as a secondary criterion. According to these cut-off values, only Polar VS demonstrated valid detection of both HR_{min} and HR_{max} and can thus be suggested for this specific application. All other optical sensors presented a difference of more than ±3 bpm for more than 20% of the participants. In any case and as a general precaution, in individuals with underlying health conditions, an unsupervised maximal exercise test should be performed after medical clearance only.

Recent recommendations suggest to assess within-device reliability by comparing the variability of average HR over fixed time windows within one activity in each participant [17]. However, we opted to investigate day-to-day reliability by repeating the same standardized laboratory-based activities on the second day. This approach adds valuable information for selecting optical sensors for PA monitoring and intervention, considering that in addition to technical error also biological day-to-day variability is acknowledged. The results indicate that upper arm-worn sensors demonstrate more consistent performance between day 1 and day 2 compared to the wrist-worn sensors. Despite the inferior reliability values observed in the wrist-worn sensors, it can be inferred that even Garmin V2, with the highest MAE of 4.6 bpm, maintains sufficient consistency for PA monitoring and recommendations.

Based on the original question of whether the optical HR sensors are valid enough for HR-based exercise monitoring and prescription we can conclude that all examined optical sensors can be recommended for this application with the exception of the Polar M2 sensor. The Polar M2 showed a MAPE of over 5% during treadmill running, which is an important finding, considering walking and running a relevant type of training modality to increase PA. The other three sensors are sufficiently valid to monitor training sessions based on HR and physiological thresholds, especially on a cycling ergometer with steady-state activities all sensors performed well. Additionally, upper arm-worn sensors constantly showed superior validity compared to wrist-worn sensors and the validity of the upper arm-worn sensors was even increasing at higher intensities on the cycle ergometer.

In terms of usability and practical applicability in medical settings, it's worth noting that upper arm-worn sensors need to be connected to another device such as a smartphone, if real-time feedback of HR is warranted. Additionally, participants rated wrist-worn optical sensors more favorably in terms of suitability for everyday use and visual appeal, attributes that can be crucial for the sensor's practical application and sustained usage.

A limitation of this study is the absence of walking and running at steady-state intensities in the validation protocol. Such exercises are commonly performed and should be implemented in future validation studies. Another limitation arises from the fact that study personnel ensured that all sensors were mounted according to the manufacturer's instructions. In outpatient settings, the self-mounting of sensors by patients may impact measurement quality. Furthermore, the classification of PA intensity zones, based on individual measurements of HR_{max}, warrants caution when applied to patients taking HR-limiting medication, especially for light and moderate intensity zones. Considering the small number of participants with darker skin tones (6 participants with Fitzpatrick 4–6) a subgroup analysis was not feasible. However, several studies have shown a higher measurement error for participants with darker skin [41].

5 | Perspective

Consumer-grade wearable devices hold promise to scale up objective device-based PA monitoring and personalized exercise intervention and thereby contribute to improving individual and public health. However, to do so, validity needs to be assessed in research organizations free of conflicting interests and using a sound methodological approach based on the most recent recommendations. This is the first study acknowledging and adhering to contemporary and comprehensive validation standards, thereby aggregating validity and reliability data that aims to facilitate healthcare professionals making informed decisions when choosing optical HR sensors for PA monitoring and intervention. Given the rapid evolution of digital health technologies, the comprehensive methodology used in this study may serve as a blueprint for future validation studies. Subsequently, wearables with high validity and reliability may serve for novel mobile health concepts such as Just-in-Time Adaptive Interventions [42] or contribute to a connected health data ecosystem to better inform personalized health decisions [43].

Acknowledgments

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Ethics Statement

The Ethics Committee of the State of Salzburg approved the study protocol, informed consent forms, and all Supporting Information study documents (No. 1017/2022).

Consent

Informed consent was obtained from all participants.

Conflicts of Interest

The authors declare no conflicts of interest.

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Data Availability Statement

The data that support the findings of this study are openly available at https://zenodo.org/doi/10.5281/zenodo.13767291.

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